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Basic science

The role and mechanism of CRISPLD2 in skin fibrosis of systemic sclerosis

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Abstract

Objectives: SSc is an autoimmune connective tissue disease involving multiple organs. The most common clinical symptom of SSc is progressive fibrosis of the skin, and the pathologically manifestations of skin were activation and proliferation of fibroblasts and continuous proliferation of extracellular matrix. TGF-β can promote the proliferation and activation of fibroblasts, causing excessive deposition of collagen and structural proteins. Therefore, exploring the specific mechanism of TGF-β-related pathway on fibrosis is of great significance for improving skin fibrosis in SSc.

Methods: Genes related to TGF-β pathway were screened through bioinformatics analysis, and SSc phenotypes were verified *in vivo* and *in vitro*. The relevant molecular mechanisms were preliminarily discussed in combination with transcriptome sequencing.

Results: Human cysteine-rich secreted protein LCCL domain protein 2 (CRISPLD2) was found increased reactivity in TGF-β-induced fibroblasts, and the expression of ACTA2 (α-SMA) decreased significantly in TGF-β-mediated fibroblasts with up-regulation of CRISPLD2.

Conclusion: CRISPLD2 was found to have increased reactivity in TGF-β-induced fibroblasts, and we further confirmed that CRISPLD2 can participate in TGF-β-induced fibroblast fibrosis from multiple perspectives and levels in negative feedback regulation, and investigated the mechanism of CRISPLD2 in fibrosis.

Keywords: SSc, CRISPLD2, TGF-β, fibrosis, fibroblasts.

Rheumatology Key messages

- CRISPLD2 is specifically highly expressed in SSc skin, which is correlated with clinical fibrosis.
- TGF-β can induce high expression of fibrosis genes, and CRISPLD2 increased reactivity in fibroblasts.
- CRISPLD2 can inhibit TGF-β-induced fibrosis in skin fibroblasts, and negative feedback is increased in SSc.

Introduction

SSc is a rare autoimmune connective tissue disease involving multiple organs, characterized by inflammatory responses, vascular changes and fibrosis, leading to extensive organ fibrosis [1]. SSc can widely affect the whole body, causing progressive fibrosis of the lung, kidney, skin and other organs. Pulmonary fibrosis is the main reason caused by death in SSc patients [2]. However, the degree of skin fibrosis is widely heterogeneous and seriously affects the quality of life of patients. Skin fibrosis often begins at the end of the limb and progresses to the proximal end [3]. The complex immune environment, including abnormally activated immune cells like macrophages and a variety of pro-inflammatory factors, interacts with stromal cells involving fibroblasts, and they jointly promote the accumulation of extracellular matrix (ECM) and the process of fibrosis [4, 5].

TGF-β is a multifunctional cytokine produced by many types of cells and regulates a wide range of cellular functions and biological processes, including angiogenesis, wound healing and embryonic development [6]. TGF-β is also an important factor in the occurrence, development and progression of fibrosis. TGF-β has been repeatedly shown to have powerful fibrosis activity *in vitro* and *in vivo* [7]. TGF-β can be involved in fibrosis by promoting fibroblast infiltration and proliferation, myofibroblast production, increased ECM production and inhibition of collagen dissolution [8]. By

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blocking the effect of TGF-β on fibroblasts, it provides a broad idea for the treatment of fibrosis diseases. Although ongoing research has focused on the development of novel TGF-β inhibitors for the prevention and treatment of fibrosis, sustained inhibition of TGF-β may cause more adverse reactions and lead to unacceptable toxicity due to its high pleotropic nature [8]. Therefore, partial inhibition of TGF-β signal conduction or targeted TGF-β signalling pathways downstream of signalling molecules may reduce the occurrence of adverse reactions.

Human cysteine-rich secreted protein LCCL domain protein 2 (CRISPLD2), also known as CAPLD2, with the protein structure consisting of a signal peptide, a CAP domain, a hinge domain and a unique LCCL domain comprising the C-terminal [9]. CRISPLD2 is shown to be associated with chronic bronchial lung dysplasia. Moreover, in human airway fibroblasts, knockdown CRISPLD2 resulted in significantly increased expression of Interleukin (IL)-8, C-C Motif Chemokine Ligand 2 (CCL2) and IL-6, suggesting that CRISPLD2 has anti-inflammatory properties [10, 11]. In the early or advanced stage of SSc, the expression level of CRISPLD2 may be closely related to the activity of inflammatory response. In this study, we focused on changes of CRISPLD2 in SSc, and further verified the function and mechanism of CRISPLD2 in fibroblasts fibrosis.

Methods

Full details of the methodology can be found in Supplementary Data S1, available at *Rheumatology* online. This study was approved by the Ethics Committee of Xiangya Hospital, Central South University (No: 201212074). All participants provided written informed consent.

Results

CRISPLD2 expression was up-regulated in multiple fibrosis-related cohorts and correlated with clinical indicators of SSc

First, we jointly analysed transcriptomic expression levels in a cohort of SSc patients, a cohort of keloids lesions and a cohort of TGF-β-stimulated fibroblasts. In these different cohorts, we found 28 co-upregulated genes that may be associated with TGF-β-mediated fibrosis and participate in the pathophysiological processes of fibrosis in SSc (Fig. 1A). Among the co-upregulated genes, collagen-related genes (*COL1A1, COL1A2*), ECM and cytoskeleton-related proteins (*ACTA2, FN1*) were also up-regulated in multiple cohorts (Fig. 1B and C). The simultaneous rise of these fibrosis-related indicators suggests that TGF-β-mediated fibroblast fibrosis participates in the pathophysiological process of SSc. Therefore, exploring the specific mechanism of TGF-β-mediated fibrosis is of great significance for understanding SSc. In addition to these fibrosis indicators, high expression of other genes such as *CRISPLD2* was also found in all three cohorts (Fig. 1B and C). The results suggest that increased *CRISPLD2* transcription levels are associated with TGF-β stimulation and may be involved in the fibrotic process of SSc.

In order to determine whether *CRISPLD2* may be involved in the occurrence and development of SSc, we comprehensively analysed the skin fibrosis status, clinical indicators and skin transcriptome expression levels of SSc. The results showed that fibrosis indicators (*COL1A1, COL1A2, ACTA2*) were positively correlated with clinical indicators mRSS score in SSc patients, and the expression level of *CRISPLD2* was also positively correlated with mRSS score and the rate of skin thickness progression (Fig. 1D). Further analysis showed that the expression of *CRISPLD2* was also positively correlated with the expressions of fibrosis indicators *COL1A1, COL1A2* and *ACTA2* (Fig. 1E). Compared with patients with diffuse cutaneous SSc (dcSSc) and localized cutaneous SSc (lcSSc), fibrosis indicators (*COL1A1, COL1A2* and *ACTA2*) and *CRISPLD2* expression was significantly higher in dcSSc patients than in healthy controls and lcSSc (Supplementary Fig. S1A, available at *Rheumatology* online). The results show that *CRISPLD2* is correlated with clinical indicators, and has the potential to become an indicator of early clinical diagnosis and prognosis.

Further analysis of single-cell transcriptome sequencing in keloid skin shows that the skin tissues were composed of 10 types of cells, including fibroblasts, myofibroblasts, endothelial cells, keratinocytes, macrophages, lymphocytes, nerve cells, pigment cells and mast cells. Fibroblasts, myofibroblasts, endothelial cells and keratinocytes were the main parts of SSc patients (Supplementary Fig. S1B and C, available at *Rheumatology* online). Compared with healthy controls, the proportion of fibroblasts and myofibroblasts increased in keloid skin (Supplementary Fig. S1D, available at *Rheumatology* online). At the same time, we found that *CRISPLD2* was mainly expressed in fibroblasts and myofibroblasts, and overlapped with the expression groups of collagen-related genes (*COL1A1, COL1A2*) (Supplementary Fig. S1E, available at *Rheumatology* online). Differences were analysed, and downstream pathway enrichment was performed by comparing transcriptome levels between *CRISPLD2*-expressing and *CRISPLD2*-non-expressing fibroblasts. Further gene ontology (GO) functional enrichment analysis of differential genes showed significant enrichment in ECM structure, wound repair, TGF-β-related signalling pathways and growth factor stimulation-related pathways (Supplementary Fig. S1F, available at *Rheumatology* online).

CRISPLD2 was increased in primary skin fibroblasts of SSc patients, mouse models of skin fibrosis and TGF-β-induced fibroblasts

Additionally, we verified the association between *CRISPLD2* and fibrosis in two kinds of mouse skin fibrosis models. Hypochlorous acid (HOCl)-induced and Bleomycin (BLM)-induced fibrosis mouse models are classical and effective animal models of SSc, and skin fibrosis can be obtained by subcutaneous injection of HOCl or BLM. We observed the skin tissue thickening and folding are reduced in the skin tissue of BLM-induced fibrosis mouse model, and *CRISPLD2* is obviously expressed on the spindle cells of the dermis (Fig. 2A). And, the expression level of *CRISPLD2* in the skin tissue of HOCl-induced fibrosis mouse model was higher than that of the control group (Fig. 2C).

Further, we examined the association between *CRISPLD2* and fibrosis in the skin of SSc patients. Comparing with normal controls, we observed that the skin tissues of SSc patients were thickened and the dermal folds were reduced, and *CRISPLD2* was significantly expressed in the fusiform cells of the dermis of SSc skin tissues (Fig. 2B). And, there was significant increase of *CRISPLD2* expression in SSc fibroblasts compared to the healthy control (Fig. 2D, Supplementary Table S1, available at *Rheumatology* online). The expression levels of primary

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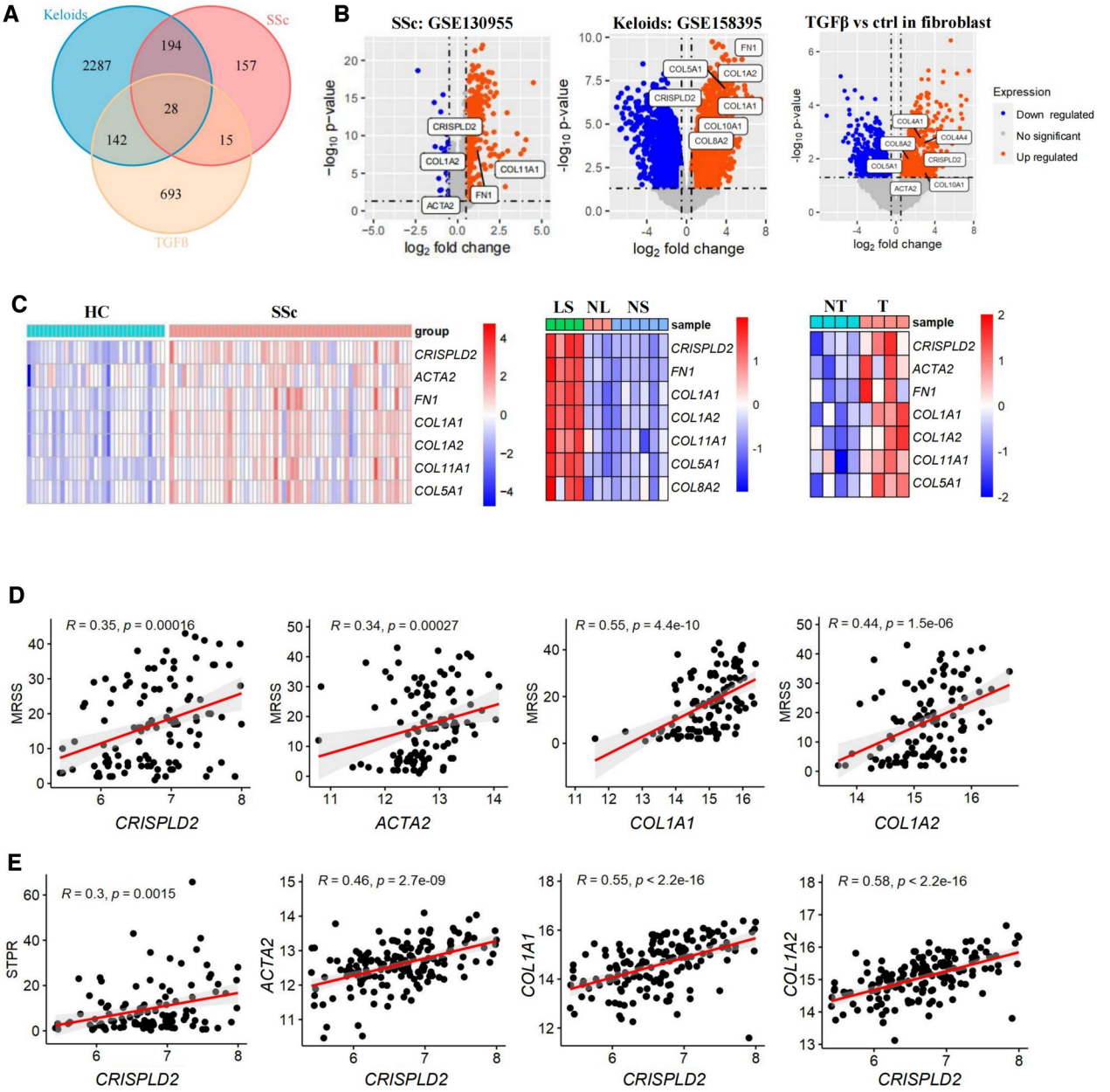

Figure 1. Expression levels of fibrosis genes and *CRISPLD2* genes in multiple cohort studies. (**A**) The Venn diagram showed co-upregulated genes present in the three cohorts. The top right circle is the number of genes upregulated in SSc, the top left circle is the number of genes upregulated in chronic keloids (Keloids) and the bottom circle is the number of genes upregulated in TGF-β-stimulated fibroblasts (TGF-β). (**B**) Volcanic maps showed some co-upregulated differential genes in the three cohorts. From left to right, the transcriptome expression levels of fibrosis genes and *CRISPLD2* in a SSc cohort, a cicatricial lesion cohort and a TGF-β-stimulated fibroblast cohort are shown. *P* value <0.05 indicates a significant difference. (**C**) Heat maps showed the transcriptome expression levels of co-upregulated genes in different individuals of the three cohorts. From left to right are the cohort of patients with SSc (HC: healthy control group, SSc: SSc group), the cohort of patients with scar lesions (NS: normal skin group, NL: normal skin surrounding scarred group, LS: scarred skin group) and the cohort of fibroblasts stimulated by TGF-β (NT: un-stimulation group, T: stimulation group). (**D**) Scatter plot showed the correlation between mRSS score, *CRISPLD2* and fibrosis gene in SSc cohort. (**E**) Scatter plot showed the correlation between *CRISPLD2*, STPR and fibrosis gene in SSc cohort.

fibroblasts in SSc skin and normal controls were analyzed, and results showed that *CRISPLD2* and fibrosis genes (*COL1A1*, *COL1A2*, *ACTA1*, *ACTA2*, *FN1*) were significantly increased in primary fibroblasts of SSc patients (Fig. 2E).

The expression levels of fibrosis genes and *CRISPLD2* in human fibroblasts were further analysed under TGF-β stimulation. The results showed that with the extension of TGF-β stimulation duration (12, 24, 48h), the fibrosis genes (*ACTA1*, *ACTA2*, *CTGF*) and *CRISPLD2* increased at different stimulation time points compared with those without stimulation (Supplementary Fig. S1G, available at *Rheumatology* online). In addition, in different concentrations of TGF-β stimulation (1, 5, 10, 20 ng/ml), the fibrosis genes (*COL1A1*, *COL1A2*, *ACTA1*, *ACTA2*, *FN1*, *CTGF*) basically increased in different stimulus concentrations compared with those without stimulation (Supplementary Fig. S1H, available at *Rheumatology* online). Combined with the duration and concentration of TGF-β stimulation, we found that the fibrosis genes increased significantly when TGF-β was stimulated at 10 ng/ml for 24–48 h. After TGF-β (10 ng/ml) stimulation for 48 h, the fibrosis genes

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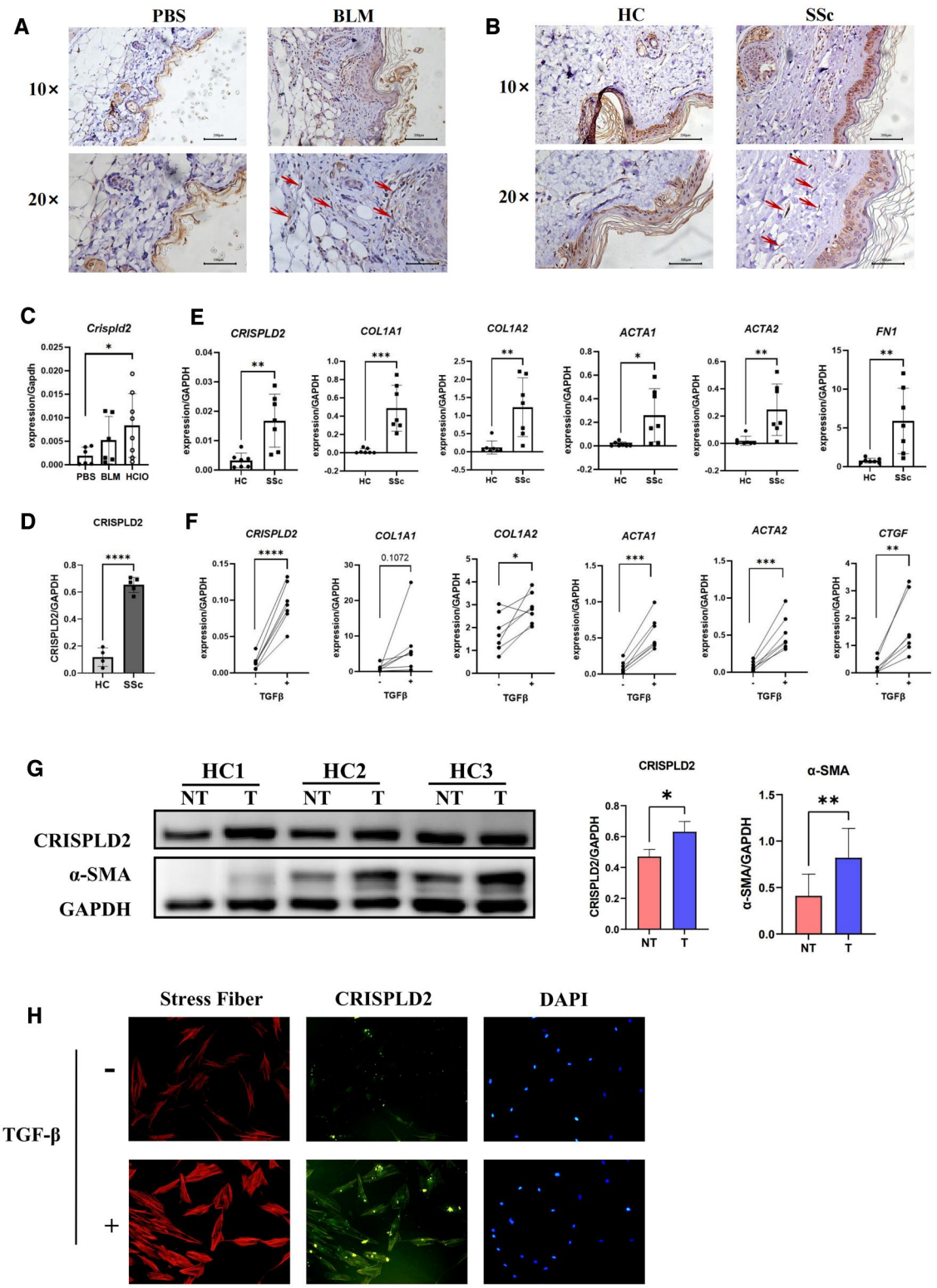

Figure 2. *CRISPLD2* was increased in primary skin fibroblasts of SSc patients, mouse models of skin fibrosis and TGF-β-induced fibroblasts. (**A**) The expression of *CRISPLD2* in skin paraffin sections of PBS and BLM was detected by IHC. The arrows indicated spindle-forming fibrocytes that are highly expressing *CRISPLD2*. (**B**) The expression of *CRISPLD2* in skin paraffin sections of normal controls and SSc was detected by IHC. The arrows indicated spindle-forming fibrocytes that are highly expressing *CRISPLD2*. (**C**) Histogram showed the difference of *CRISPLD2* mRNA expression levels in PBS, BLM-induced and HOCl–induced fibrosis mouse models. (**D**) Histogram showed the relative protein expression level of *CRISPLD2* and α-SMA. *P* value < 0.0001 (****). (**E**) Histograms showed the difference of *CRISPLD2* and fibrosis gene expression levels in primary fibroblasts of HC and SSc. (**F**) Scatter diagram showed the changes in mRNA expression levels of fibrosis gene and *CRISPLD2* in seven primary fibroblasts in the TGF-β stimulated group (10 ng/ml, 48 h) compared with that in the TGF-β unstimulated group. Paired *t*-test was used, *P* value< 0.05 indicates a significant difference, where *P* value < 0.05 (*), *P* value < 0.01 (**), *P* value < 0.001 (***), *P* value < 0.0001 (****). (**G**) WB verification showed that the protein levels of α-SMA and CRISPLD2 genes in the six primary fibroblasts in the TGF-β stimulated group (10 ng/ml, 48 h) were different compared with those in the TGF-β unstimulated group. Histogram showed the relative expression level of *CRISPLD2* and α-SMA. *P* value < 0.05 (*), *P* value < 0.01 (**). (**H**) The levels of cytoskeleton protein stress fibre and *CRISPLD2* gene in primary fibroblasts in the TGF-β stimulated group (10 ng/ml, 48h) were verified by cellular immunofluorescence compared with those in the TGF-β unstimulated group.

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(*COL1A1, COL1A2, ACTA1, ACTA2, CTGF*) and *CRISPLD2* in multiple primary fibroblasts were significantly increased compared with those in the unstimulated group (Fig. 2F). Moreover, results of western blotting also verified that the protein levels of *CRISPLD2* and α-SMA were significantly increased in TGF-β stimulated group compared with those unstimulated (Fig. 2G), and the expression of α-SMA and *CRISPLD2* was positively correlated without interference of *CRISPLD2* (Supplementary Fig. S1I, available at *Rheumatology* online). Protein spectrum analysis showed that the protein levels of *CRISPLD2* and fibrosis genes were significantly increased after TGF-β stimulation for 48 h (Supplementary Fig. S2A and B, available at *Rheumatology* online). In order to visually demonstrate the change of fibrosis level and the increase of *CRISPLD2* gene expression under TGF-β stimulation, we performed cellular immunofluorescence on cultured fibroblasts after stimulation. The results showed that the expression of *CRISPLD2* gene in fibroblasts was significantly increased under TGF-β stimulation, and the cell size was slightly increased, and the cytoskeleton was significantly increased and enhanced under TGF-β stimulation (Fig. 2H).

Knocking down *CRISPLD2* can up-regulate TGF-β induced fibroblast fibrosis

Previously, we analysed and validated the *CRISPLD2* increased in TGF-β-induced fibroblast fibrosis. We further examined the effect of *CRISPLD2* on fibrosis genes by knocking down the expression of *CRISPLD2* in fibroblasts. The results showed that TGF-β stimulated the expression levels of fibroblasts fibrosis genes such as *ACTA1, ACTA2, FN1* and *COL1A2*, and the expression levels of fibrosis genes significantly increased after *CRISPLD2* gene knockdown (Fig. 3A). And the expression level of α-SMA increased in siCRISPLD2-T compared with siNC-T group (Supplementary Fig. S2C, available at *Rheumatology* online). Cytoskeleton of fibroblasts increased significantly after *CRISPLD2* knockdown was verified by cell fluorescence assay (Fig. 3B). Further, five primary fibroblasts were treated with siNC-NT, siNC-T, siCRISPLD2-NT and siCRISPLD2-T, and RNA-seq was performed. First, principal component analysis (PCA) was performed that the four different treatments were better clustered together (Fig. 3C). Analysis showed that the expression levels of *CRISPLD2, ACTA2* (α-SMA), *TGFBR1* and other fibrosis-related genes increased in siCRISPLD2-T group (Fig. 3D, Supplementary Fig. S2D, available at *Rheumatology* online). GSEA functional enrichment analysis mainly concentrated on cell cycle-related pathway, DNA replication-related pathway, nuclear membrane transport-related pathway and calcium ion channel correlation (Fig. 3E). Furthermore, we found that some genes (*CDC20, MKI67, CDCA5*) related to cell cycle were significantly increased in siCRISPLD2-T group than siNC-T group (Supplementary Fig. S2H and I, available at *Rheumatology* online).

Up-regulation of *CRISPLD2* can inhibit TGF-β induced fibroblast fibrosis

Our analysis confirmed that *CRISPLD2* knockdown promotes fibroblasts fibrosis. Therefore, in order to further investigate the effect of *CRISPLD2* on fibrosis, we detected its effect on fibrosis genes by up-regulating *CRISPLD2* expression in fibroblasts. The results showed that TGF-β stimulated the expression levels of fibroblasts fibrosis genes such as *ACTA1, ACTA2, FN1* and *COL1A2* decreased after the up-regulation of *CRISPLD2*, especially the significant decrease of *ACTA1* and *ACTA2* (Fig. 4A). The expression level of ACTA2 (α-SMA) protein in pcCRISPLD2-T decreased compared with pcNC-T with unknockout (Supplementary Fig. S2E, available at *Rheumatology* online). Meanwhile, the expression of α-SMA and *CRISPLD2* was negatively correlated with interference of *CRISPLD2* (Supplementary Fig. S2G, available at *Rheumatology* online). The cytoskeleton of fibroblasts was significantly reduced after *CRISPLD2* upregulation by cell fluorescence assay (Fig. 4B). Further, five primary fibroblasts cell lines were treated with pcNC-NT, pcNC-T, pcCRISPLD2-NT and pcCRISPLD2-T, and RNA-seq was performed. First, PCA analysis was performed that the four different treatment methods were better clustered together (Fig. 4C). Analysis showed that the expression levels of *CRISPLD2* and *ACTA2* (α-SMA) genes were down-regulated in pcCRISPLD2-T group (Fig. 4D, Supplementary Fig. S2F, available at *Rheumatology* online). GSEA functional enrichment analysis mainly concentrated on cytokine and receptor interaction, IL-17 signalling pathway, TNF signalling pathway, Toll-like receptor signalling pathway, JAK-STAT signalling pathway, MAPK signalling pathway and so on (Fig. 4E).

Discussion

Fibroblasts stimulated by TGF-β shows cell proliferation, increased collagen expression and gradually differentiated into myofibroblasts. Therefore, inhibiting the stimulation of TGF-β on fibroblasts can effectively slow down the process of fibrosis. Because TGF-β in the human body can not only promote fibroblast fibrosis, but also mediate angiogenesis, immunochemotaxis and other effects [7]. Further analysis of TGF-β through which downstream molecules or pathways can accelerate the develpment of targeted fibrosis therapy. We found that the anti-inflammatory protein CRISPLD2 is upregulated in TGF-β-stimulated fibroblasts as an indicator of fibrosis and is specifically expressed in fibroblasts. With CRISPLD2 knocking down, TGF-β-mediated fibroblast fibrosis level is increased, and when CRISPLD2 is up-regulated, TGF-β-mediated fibroblast fibrosis level is decreased. Therefore, the reactivity of CRISPLD2 in TGF-β-mediated fibroblasts was increased, and the expression of CRISPLD2 in high concentration had a certain inhibitory effect on TGF-β-mediated fibrosis.

TGF-β can mediate cell cycle, inflammation and apoptosis to induce fibrosis, especially the differentiation of cell cycle activity of different cell types is closely related to fibrosis in multiple organs [12]. Fibroblasts provide essential niche and location information to neighbouring cells through the regulated secretion of microstructural, biomechanical and biochemical cues in ECM, as well as soluble mediators such as cytokines, growth factors and metabolites [13, 14]. In an injured or inflammatory state, fibroblasts are activated and over-proliferate, transforming into myofibroblasts, which lead to fibrosis through the continuous accumulation of ECM. Among them, severe or repeated injuries or inflammatory states may lead to cell cycle disorders, such as cell cycle arrest in epithelial cells and over-proliferation of fibroblasts, both of which are closely associated with fibrosis [15]. Cell cycle arrest is closely connected with apoptosis, so it has been found that the degree of fibrosis can be significantly improved by inducing the cell cycle, such as inhibiting the cell cycle progression of fibroblasts or myofibroblasts can significantly improve fibrosis and cell hyperproliferation [16]. Although cell cycle regulators play an important role in fibrosis,

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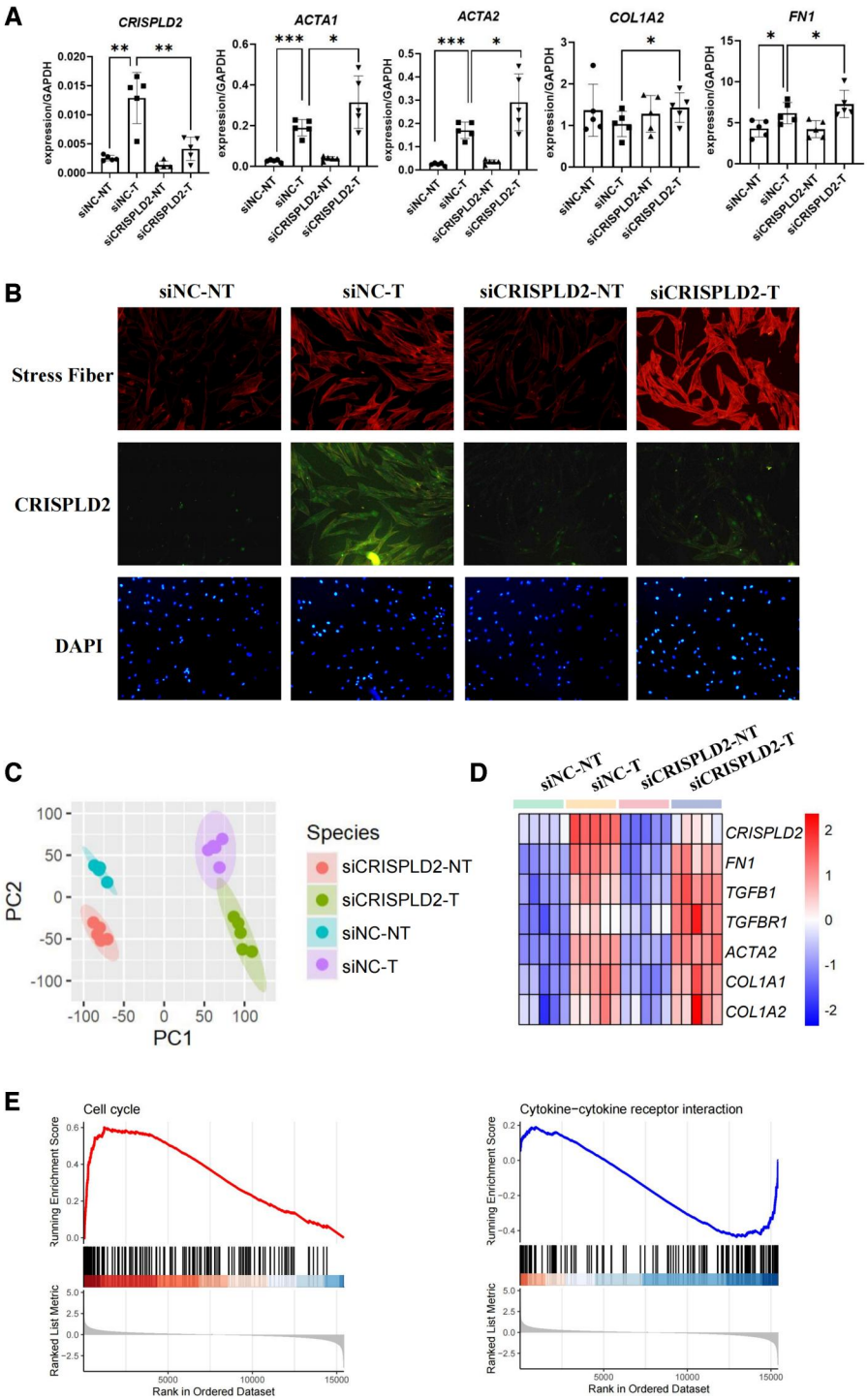

Figure 3. Knocking down *CRISPLD2* can up-regulate TGF-β-induced fibroblast fibrosis. (**A**) Bar chart showed the mRNA expression levels of fibrosis gene and *CRISPLD2* in five primary fibroblasts treated by siNC-NT, si-NC-T, si-CRISPLD2-NT and siCRISPLD2-T. Using ANOVA test, P value < 0.05 (*), P value < 0.01 (**), P value < 0.001 (***). (**B**) Cellular immunofluorescence verified the levels of cytoskeletal protein stress fibre and *CRISPLD2* gene in fibroblasts cells under four different treatments: siNC-NT, siNC-T, siCRISPLD2-NT and siCRISPLD2-T. (**C**) PCA analysis was performed on RNA-seq sequencing samples (SICRINT: siCRISPLD2-NT, SICRIT: siCRISPLD2-T, SINCNT: siNC-NT, SINCT: siNC-T). (**D**) Heat maps showed the expression levels of *CRISPLD2* and fibrosis genes under four different treatments. (**E**) Visualization of GSEA results showed enrichment of GSEA pathway of differential genes.

there are few studies on cell cycle regulators in fibrosis. Most of the studied regulators are CKI-related molecules, and there is very little information on other cell cycle regulators. The regulatory changes of CRISPLD2 are related to cell cycle and process changes, and CRISPLD2 may mediate tissue fibrosis by affecting the cell cycle of fibroblasts with the specific mechanism remains to be further verified [15]. In SSc, the activation of endothelial and immune cells leads to chronic inflammation, which in turn activates fibroblast-dependent production of excess reactive oxygen species, leading to abnormal wound

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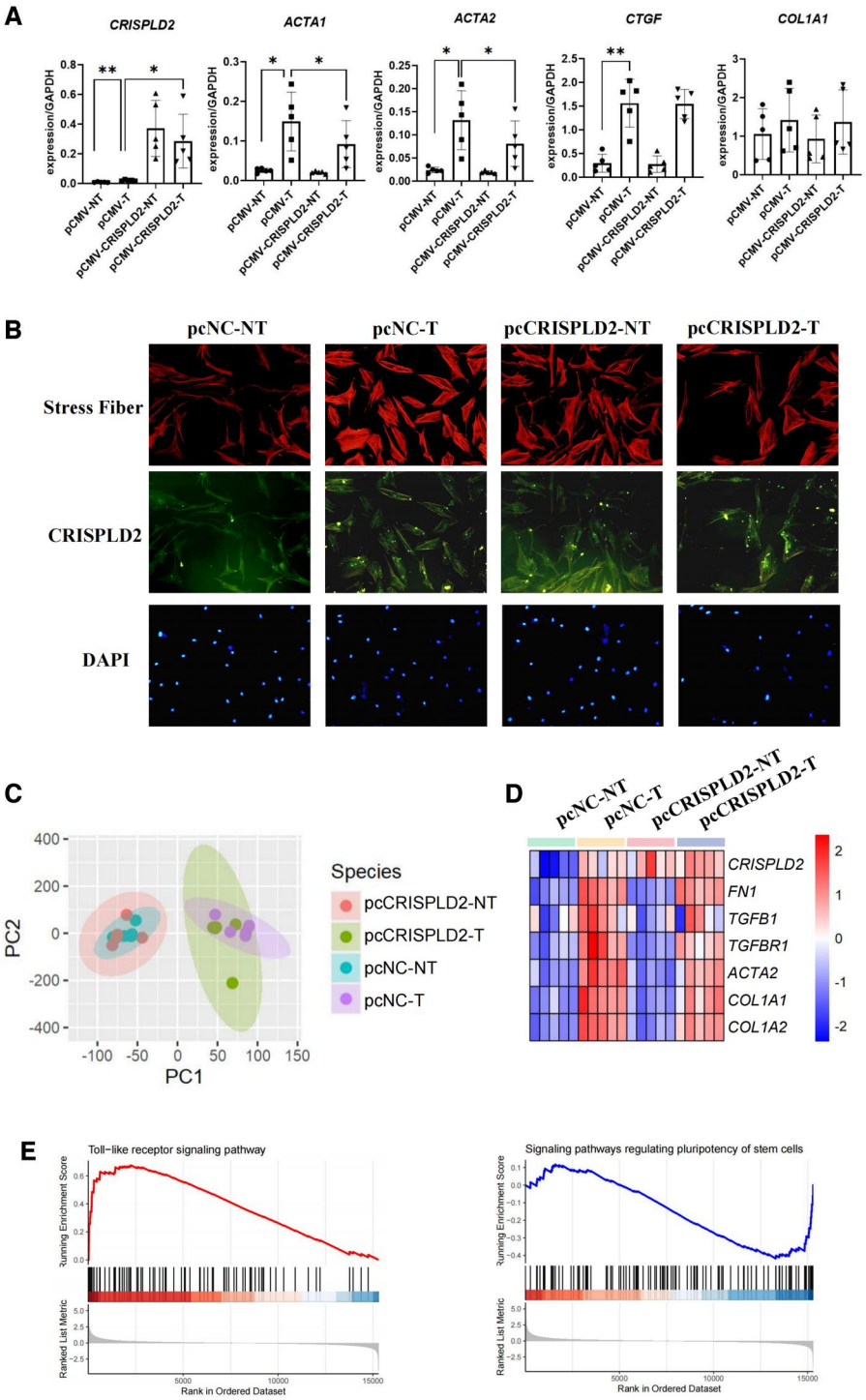

Figure 4. Up-regulation of *CRISPLD2* can improve TGF-β-induced fibroblast fibrosis (**A**) Bar chart showed the mRNA expression levels of fibrosis gene and *CRISPLD2* in five primary fibroblasts treated with four different treatments: pcNC-NT, pcNC-T, pcCRISPLD2-NT and pcCRISPLD2-T. Using ANOVA test, P value < 0.05 (*). (**B**) Cellular immunofluorescence verified the levels of cytoskeletal protein stress fibre and *CRISPLD2* in fibroblasts under four different treatments: pcNC-NT, pcNC-T, pcCRISPLD2-NT and pcCRISPLD2-T. (**C**) PCA analysis was performed on RNA-seq sequencing samples (PCCRINT: pcCRISPLD2-NT, PCCRIT: pcCRISPLD2-T, PCNCNT: pcNC-NT, PCNCT: pcNC-T). (**D**) Heat maps showed the expression of *CRISPLD2* and fibrosis genes under four different treatments. (**E**) Visualization of GSEA results showed enrichment of GSEA pathway of differential genes.

healing and fibrosis of the skin and internal organs. Excess ROS can be activated by MAPK and incidentally regulate the TGF-β pathway through ERK-mediated phosphorylation of SMAD [17]. TGF-β increased the production of reactive oxygen species by increasing the induction of NADPH oxidase and inhibiting the synthesis of glutathione and other mitochondrial function impairment, which led to the subsequent aggravation of oxidative stress. Thus, the interaction between TGF-β and ROS in SSc is a vicious cycle that maintains and intensifies the fibrotic process [18]. The regulatory changes of CRISPLD2 are related to oxidative stress, so CRISPLD2 may be involved in regulating TGF-β-mediated tissue fibrosis by influencing the production of

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reactive oxygen species. In the process of fibrosis, a variety of cytokines affect to fibroblasts. Despite the complexity of cytokine networks, pro-fibrotic cytokines appear to have a degree of specialization, with some affecting collagen production and others proliferating. For example, TNF-α tends to stimulate proliferation, while TGF-β tends to stimulate collagen production, and IL-4 can up-regulate both collagen production and proliferation [19]. During TGF-β-mediated fibrosis, activation of toll-like receptor signalling pathways can induce an enhanced TGF-β response, which in turn modulates collagen production and enhanced ECM deposition [20]. CRISPLD2 affects cytokines interaction with receptors, such as IL-17 signalling pathways, TNF signalling pathways, toll-like receptors signalling pathways, JAK-STAT signal pathway, MAPK signalling pathways, and other signalling pathways. TGF-β-mediated fibrosis may be influenced through these signalling pathways by modulating CRISPLD2 expression.

However, it remains to be further verified that CRISPLD2 intervenes in TGF-β-mediated fibroblast fibrosis by affecting the degree of cell cycle activation and oxidative stress. In addition, whether CRISPLD2 intervention in TGF-β-mediated fibroblast fibrosis can affect other biological functions of TGF-β remains to be studied, and more data and experimental support for clinical diagnosis and treatment through observation and regulation of CRISPLD2 changes are still needed.

Conclusion

In this study, CRISPLD2 was found increased reactivity in TGF-β-induced fibroblasts, and we further confirmed that CRISPLD2 can participate in TGF-β-induced fibroblast fibrosis from multiple perspectives and levels, and CRISPLD2 may inhibit TGF-β-induced fibroblast fibrosis through a negative feedback mechanism. This provides evidence for CRISPLD2 to become a clinical potential therapeutic target and a new biomarker.

Supplementary material

Supplementary material is available at *Rheumatology* online.

Data availability

RNA-seq data were downloaded from the Gene Expression Omnibus (GEO) database, including SSc cohort (GSE130955), SSc clinical cohort (GSE181549) and keloid skin (GSE158395). scRNA-seq data of SSc skin (GSE163973) were also downloaded from the GEO database. The data of fibroblasts transcriptome sequencing (GSE264288) was uploaded to GEO database. The data of cohort of TGF-β-stimulated fibroblasts are available from the corresponding author on reasonable request.

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